Department of Mathematics and Applied Mathematics

Module 3MS: Metric Spaces 2023

Class test 2 solutions **Total marks: 40**

1. (a) $([0,1], d_E)$ (b) $(\mathbb{R}, d_E)$ (c) Impossible (d) $(\mathbb{Q}, d_E)$
(e) Any function $f : X \to Y$ from discrete metric space to any metric space is continuous. That is, if G is open Y, then preimage $f^{-1}(G)$ is open X. In particular, our function in question is continuous.

[2×5=10]

2. Suppose that U is an open subset of (Y, d_Y).
Then $Y \setminus U = U'$ is a closed subset of (Y, d_Y).
By assumption, $f^{-1}(U')$ is a closed subset of (X, d_X).
But $f^{-1}(U') = (f^{-1}(U))'$, so $f^{-1}(U)$ is an open subset of (X, d_X).

The property of inverse images used is: $f^{-1}(B') = (f^{-1}(B))'$ for any subset B of Y.
Proof: Let $x \in X$. Then:
$x \in f^{-1}(B') \iff f(x) \in B' \iff f(x) \notin B \iff x \notin f^{-1}(B) \iff x \in (f^{-1}(B))'$

[4+3=7]

3. (a) Suppose that $x_n \to x$ in a metric space (X, d). Let $\epsilon > 0$, then there exists $N \in \mathbb{N}$ such that, for any $n \in \mathbb{N}$, $n \geq N \implies d(x_n, x) < \frac{\epsilon}{2}$, so, for any $n, m \in \mathbb{N}$, $m \geq n \geq N \implies d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$.
(b) True.
Let A be a closed subset of a complete metric space (X, d).
Suppose (a_n) is a Cauchy sequence in A.
Then (a_n) is a Cauchy sequence in X.
Since X is complete, $a_n \to x$ for some $x \in X$.
Since A is closed, $x \in A$, as required.
(c) False.
Counter-example: $(\mathbb{R}, d_E)$ is a complete space; $A = (0, 2)$ is an open subset of $\mathbb{R}$ and $(x_n) = (\frac{1}{n})$ is a Cauchy sequence in A which does not converge to a point of A.

[2+2+3=7]

4. (a) Let $\epsilon > 0$ be given. Choose $N \in \mathbb{N}$ such that $n \geq N$ implies that diam $A_n < \epsilon$. Then $m \geq n \geq N$ implies that $d(x_n, x_m) \leq \mathrm{diam} A_N < \epsilon$. This follows because the sequence (A_n) is decreasing, so $x_n, x_m \in A_N$.

(b) Since (X, d) is complete, every Cauchy sequence in it converges.

(c) We have $x_n \in A_n$ and, for $n \geq k$, $A_n \subseteq A_k$. Using $B(x; r) \cap A_k = \emptyset$ we get that for $n \geq k$, $x_n \notin B(x; r)$.

(d) Since $x_n \to x$, for each $r > 0$, there exists $M \in \mathbb{N}$ such that $n \geq M$ implies that $x_n \in B(x; r)$. The statement in line (6) contradicts this.

(e) The proof from line (4) to line (7) is a proof by contradiction. The contradiction in line (7) shows that the statement in line (4) was false. That means that $x \in \overline{A_n}$ for all $n \in \mathbb{N}$; that is, $x \in \bigcap_{n\in\mathbb{N}} \overline{A}_n$.

(f) The statement is false. Let $(\mathbb{R}, d_E)$ be the usual complete metric space, take $A_n = (-\infty, -n)$. Then (A_n) is a decreasing sequence of non-empty subsets of $\mathbb{R}$. However, $\bigcap_{n\in\mathbb{N}} \overline{A}_n = \bigcap_{n\in\mathbb{N}}(-\infty, -n] = \emptyset$. [2+2+2+2+2+2=12]

5. (a) This topology $\mathcal{T}$ does not satisfy the Hausdorff property, and so is not induced by a metric: 0 and 1 are distinct real numbers, but every open set containing 0 also contains 2023; similarly every open set containing 1 also contains 2023. So there are no disjoint open sets containing 0 and 1, respectively.

(b) Suppose $f : (\mathbb{R}, d_E) \to (\mathbb{R}, d_E)$ is such that $f(n) = n^2$ for each natural number n.
For each n, $(n + 1)^2 - n^2 = (n^2 + 2n + 1) - n^2 = 2n + 1$.
So, for each n, divide the interval $[n, n + 1]$ up into $2n + 1$ equal sub-intervals.
There will exist one sub-interval, let us call it $[x_n, y_n]$, for which $|f(x_n) - f(y_n)| \geq 1$.
But $|x_n - y_n| \leq \dfrac{1}{2n+1}$.

Putting this together shows that (x_n) and (y_n) are neighbours, but $(f(x_n))$ and $(f(y_n))$ are not; so f is not uniformly continuous.

[2+2=4]